



Aishu R 251001007-2025-IT ▾

A2**Started on** Monday, 24 August 2026, 12:20 PM**State** Finished**Completed on** Monday, 24 August 2026, 12:25 PM**Time taken** 5 mins 6 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00**Problem Statement**

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

Answer: (penalty regime: 0 %)

```
1  #include <stdio.h>
2
3  int countZero(int a[], int low, int high)
4  {
5      int mid;
6
7      if (low > high)
8          return 0;
9
10     if (a[low] == 0)
11         return high - low + 1;
12
13     if (a[high] == 1)
14         return 0;
15
16     mid = (low + high) / 2;
17
18     return countZero(a, low, mid) + countZero(a, mid + 1, high);
19 }
20
21 int main()
22 {
23     int m, i;
24     int a[1000];
25
26     scanf("%d", &m);
27
28     for (i = 0; i < m; i++)
29         scanf("%d", &a[i]);
30
31     printf("%d", countZero(a, 0, m - 1));
32
33     return 0;
34 }
35
36
```



	Input	Expected	Got	
✓	5 1 1 1 0 0	2	2	✓
✓	10 1 1 1 1 1 1 1 1 1 1 1 1	0	0	✓
✓	8 0 0 0 0 0 0 0 0 0	8	8	✓
✓	17 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0	2	2	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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